

Space or Schema? Rhythm, Meter, and Pitch in C. P. E. Bach's Symphony in D Major, Wq. 183/1

ABSTRACT

This article analyzes a single piece: I discuss manipulations of rhythm, meter, harmony, and galant-schemata-within-form in C. P. E. Bach's Symphony in D major, Wq. 183/1 (of 1775 or 1776). I argue that Bach manipulates both meter finding and key finding, creating harmonic "rational deceptions" (*vernünftige Betrügereyen*) as he explicitly suggests in his instructions for improvising a keyboard fantasy (Bach 1762). In addition to creating a tonal disruption, Bach creates a meter finding problem that surprisingly resonates with nascent metric theories from his milieu around the time of composition (Kirnberger and Schulz 1774, Mirka 2009). I offer an ad hoc, multi-parametric analysis of this idiosyncratic piece, engaging its metric, tonal, schematic, and formal manipulations: I reflect on processes moving from ambiguity to clarity as well as on those shaping up a parametric grid into a recognizable schema.

1. *Rational Tonal Deceptions*

The opening of C. P. E. Bach's extraordinary Symphony in D major, Wq. 183/1, presents an unusual chain of events: a metrically ambiguous and tonally clear passage gives way to another passage that is metrically clear and tonally ambiguous.¹ When preceding ambiguities subside, the music is funneled through a highly conventional path of schemata into a familiar signpost—a Medial Caesura (Hepokoski and Darcy 2006). The modest goal of this vignette-style article is to provide a deep analytical dive into a single piece rather than to make a sweeping theoretical claim. Nevertheless, this is an opportunity to present an ad-hoc, multi-parametric analysis of an idiosyncratic piece. I will argue that the piece manipulates harmony, meter, and pitch space in

¹ The opening of Bach's symphony is analyzed briefly in Yust (2018, 187–8).

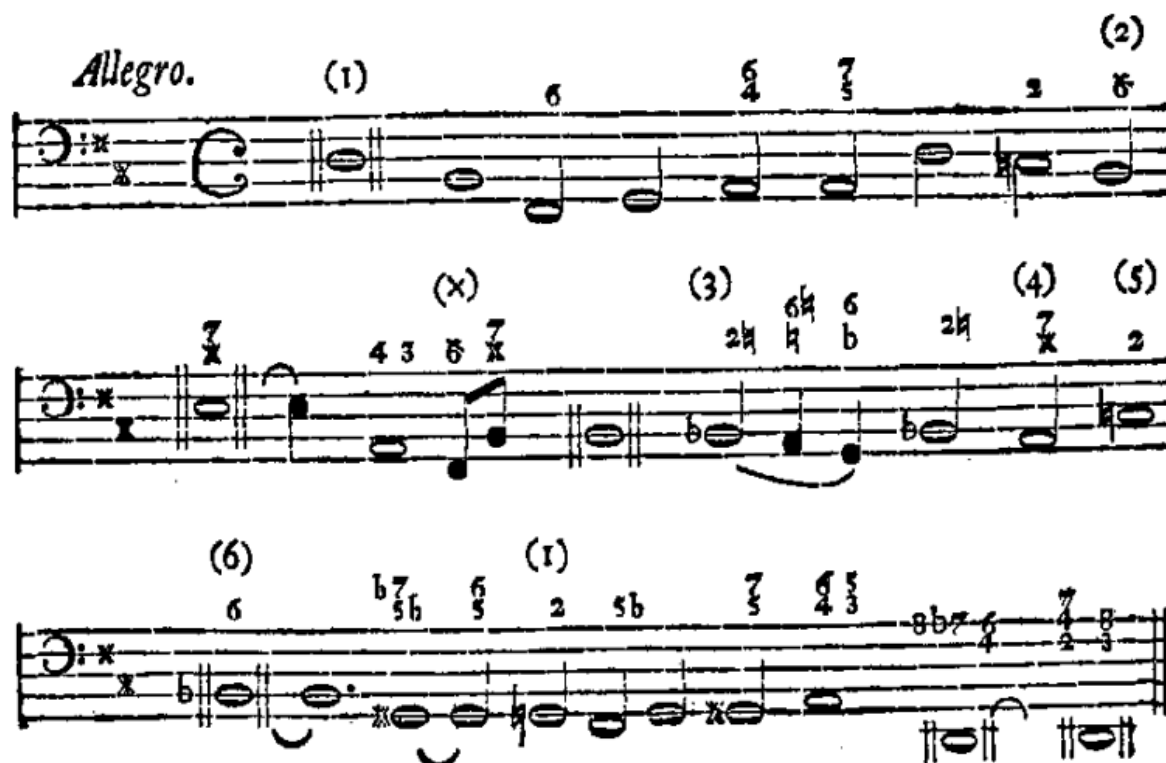
unusual ways, which invites a close reading. Based on Bach's own views on harmonic deceptions in an improvised fantasy in the second part of his treatise and on emerging metric theories, there is even some basis for reading the piece as a kind of implicit theoretical statement about equivalence between tonal and metric manipulations. Though arguments regarding authorial intent can get thorny, I suspect that Bach planted manipulations in both the pitch and meter domain.² Note that my main goal here is to present a present-day analytical hearing of the piece rather than to engage in historically informed listening. Nevertheless, the proximity in time between Bach's composition and emerging theories of meter is noteworthy.

Bach famously commented on harmonic deceptions at the concluding chapter of the second volume of his keyboard treatise (1762), which deals with improvising a fantasy. Bach explains that one can create expectations of moving in a certain tonal direction and establishing it through a formal cadence, then frustrating those expectations. Bach's outline is given in Example 1. Bach writes (in William Mitchell's translation): "It is one of the beauties of improvisation to feign modulation to a new key through a formal cadence and then to move off in another direction. This and other rational deceptions make a fantasy attractive; but they must not be excessively used, or natural relationships will become hopelessly buried beneath them."³ In thinking about harmonic deceptions in the Symphony Wq. 183/1, it is useful to consider the composer's thoughts about

² For an entry point into discussions of authorial intent, see Haimo (1996).

³ Cited in Brown (2010, 11). In the original: "Es ist bey dem Fantasiren eine Schönheit, wenn man sich stellet, durch eine förmliche Schlußcadenz in eine andere Tonart auszuweichen, und hernach eine andere Wendung nimmt. Diese, und andere vernünftige Betrügereyen machen eine Fantasie gut: allein sie müssen nicht immer vorkommen, damit das Natürliche nicht ganz und gar darbey verstecket werde." (Bach 1762, 330).

an equilibrium of surprises and “natural” relations: though we might think nowadays about harmonic conventions as cultural and style-specific, not natural, Bach suggests maintaining a balance between the two; in Wq. 183/1, this balance is maintained in both harmony and meter.



Example 1. Bach’s model fantasy outline (1762, 341).

Example 2 is a harmonic reduction (not to metric scale) of mm. 1–26 of the first movement of the symphony, summarizing the main tonal events. As we can see, after a clear I-vi-IV opening—or, if you will, a galant Romanesca schema (Gjerdingen 2007) in the home key of G major—mm. 19–20 unexpectedly shift to A minor, mm. 21–22 tonicize B minor (the key of iii) while mm. 23–26 tonicize ii, which ends up serving as ii6 in the main tonality of D major. One might therefore think about mm. 19–23 as a parenthetical, ambiguous insert into the clear opening, which disrupts a 5–6

contrapuntal motion above $\hat{4}$, from IV into ii6.

Measures 19 and 20: Treble clef has triads (D4, E4, F#4) and (D4, E4, G4). Bass clef has notes D3, E3, F#3. Measure 19 bass has a 5 below F#3. Measure 20 bass has 7b and 5 below D4, and 6b and 4 below E4.

Measures 21-26: Treble clef has triads (A4, B4, C#4), (A4, B4, C#4), (A4, B4, C#4), and (A4, B4, C#4). Bass clef has notes D3, E3, F#3, D3, E3, F#3, D3, E3, F#3, D3, E3, F#3. Measure 21 bass has 7 and 5 below D3. Measure 22 bass has a 5 below E3. Measure 23 bass has 4+ below F#3. Measure 24-26 bass has a 6 below D3.

Example 2. Harmonic reduction (not to scale) of C. P. E. Bach, Symphony in D major, Wq 183/1, mm. 1–26.

The turn of mm. 19–20 is tonally disorienting: within the realm of eighteenth-century harmonic usage, it seems to me that a fully diminished seventh chord followed by a 6/4 sonority a half step above in the bass provide unambiguous tonal position finding.⁴ Yet a cadence in A-minor fails to materialize, despite the schematic expectation that m. 20 act as a cadential 6/4 to be followed by 5/3 (or 7/5/3) and a tonic chord.

Following the ii6, the music re-orientes us towards the home key through V4/2–I resolutions (or instances of the galant *Passo Indietro* schema) in that key. The tonal clarity at this juncture is

⁴ *Position finding* is used here abstractly after Browne (1981).

subsequently reinforced through a well-worn succession of schemata (Gjerdingen 2007), a Prinner→HC combination that culminates in a I: HC MC in Hepokoski and Darcy's (2006) terms.⁵ The game of clarity and ambiguity thus encompasses harmony, schemata, and cadential conventions: the piece starts clearly, introduces ambiguity and tonal deception, and returns to clarity both harmonically and in terms of schemata and cadence points. Example 3 summarizes the tonal and schematic events until the MC, removing the blurry parentheses and highlighting the disrupted 5–6 motion above $\hat{4}$.

The image displays two systems of musical notation, each consisting of a grand staff (treble and bass clefs) with chords and fingerings indicated below the notes.

System 1:

- Romanesca (DESC 3rds):** Four chords. Bass line notes: 1, 6, 4, 5-----6. Treble line notes: 1, 6, 4, 5-----6.
- Passo Indietro *2:** Four chords. Bass line notes: 4+, 6, 4+, 6. Treble line notes: 7, 1, 7, 1.

System 2:

- Prinner:** Four chords. Bass line notes: 4, 3, 2, 1. Treble line notes: 6, 5, 4, 3.
- HC (I: HC MC):** One chord. Bass line note: 5. Treble line note: 2.

Example 3. Reductive and schematic analysis of mm. 1–34 (not to scale), omitting the tonal deception.

It may not be a coincidence that the symphony is in the key of D major, like Bach's model fantasy in his treatise. In addition, both pieces are based on an opening bass outline of (disrupted) D-B-G-F#

⁵ Regarding the cadential possibilities associated with Gjerdingen's (2007) Prinner, see Caplin (2015).

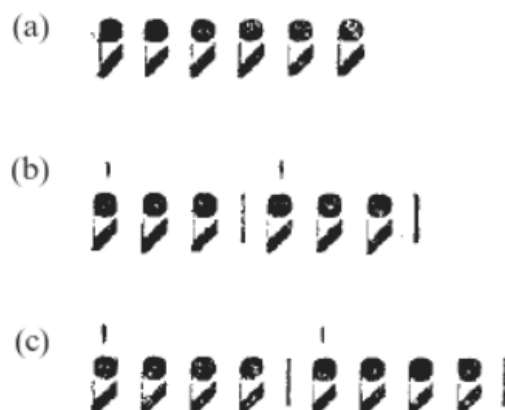
or D-B-F#-G. More significant, of course, is the fact that the symphony engages in a fantasy-like tonal deception, among other manipulations.

2. *Space vs. Schema*

After the tonally challenging opening, Bach completes his exposition in a more orderly harmonic and tonal fashion. Yet the piece tells an interesting story in the domain of rhythm and meter as it interacts with up-and-down gestures in pitch space. In so doing, Bach may be reacting to Kirnberger and Schulz's metric metaphors in their article on rhythm in Sulzer's encyclopedia, first included in volume two of the original edition, published in 1774: an artisan striking repeatedly, such as a cooper making a barrel, imposes a binary or ternary organization on a series of otherwise undifferentiated pulses. As Mirka surveys (2009, e.g., 13–14), the idea that regularity on two levels—a pulsation level and a slower one—is crucial for establishing meter, was foundational to the development of both late eighteenth-century and late twentieth-century metric theories. Kirnberger and Schulz write: “The cooper who drives a hoop [onto a barrel], [or] the coppersmith who hammers a kettle soon takes it into his head to perform his beats not singly in a completely identical way [as in Example 2/11a]; [rather] he will soon beat them [as in Examples 2.11b or 2.11c] in order to slightly differentiate the strength or the pitch [*Ton*] of the three or four beats going on one measure, so that the division of groups may become perceptible to the ear.” (Mirka's translation, 2009, 47).⁶

⁶ In the original: “Der Faßbinder, oder Böttcher, der einen Reifen antreibt, der Kupferschmied, der einen Kessel hämmert, fällt gar bald darauf, seine Schläge nicht einzeln in völliger Gleichheit so zu thun [Example 2.10a] u. s. f. er wird bald so schlagen: [Example 2.10b] u. s. f. oder so: [Example 2.10c] u. s. f. um die Stärke oder den Ton der drey, oder vier auf einen Takt gehenden Schlägen etwas abzuändern, damit die Eintheilung in Glieder dem Ohr merklich werde.” (Kirnberger and Schulz 1794, 97, cited in Mirka 2009, 47). In the second volume of the first edition, this quotation and the notations cited here as Example 4 (Mirka's example 2.11) are at Kirnberger and Schulz (1774, 980).

Example 2.11. Kirnberger and Schulz, “Rhythmus,” 97.



Example 4. Mirka (2009, 47, example 2.11), from Kirnberger and Schulz (1794, 97) [cf. 1774, 980].

According to the editorial introduction in the volume of symphonies that includes Wq 183 in the collective edition of Bach’s works, the composer rehearsed all four symphonies in August of 1776 and it is hypothesized that he may have composed the first pair in fall 1775 and the latter two in spring 1776.⁷ In other words, even if Bach could not read the discussions of rhythm and meter in Volume 2, Part 1 of Kirnberger’s (1776) *Art of Strict Musical Composition* (1982 [1776]) prior to composing the first two symphonies of Wq. 183 in (presumably) 1775, he would have been able to encounter such ideas about pulsations in Sulzer’s encyclopedia.⁸ At any rate, the proximity in time

This is the edition that would have been available to Bach while working on the symphony. See also the discussion of subjective rhythmization in London (2012, 13–15).

⁷ Kidger (2005, vii).

⁸ Kirnberger (1982 [1776], 394) writes about a C. P. E. Bach symphony in 3/2 that presents an unusual subdivision into continuous sixteenth notes. The English translators identify this as an E-flat major symphony for two horns, two oboes, two violins, viola, and bass of 1757 (*Ibid.*, fn. i). Using the recent critical edition of Bach’s works (available online), it is now easy to identify the work clearly as Wq. 179 and peruse its score: like Wq. 183/1, Wq. 179 starts with arpeggiated

and place between Bach's symphony and emerging metric theories is intriguing, as is the fact that Bach the practitioner wrote a piece that creates rational deceptions in the domains of both pitch and meter in its fantasy-like opening.

Example 5 presents the musical surface of the opening without beams and barlines. In so doing, I have tried to recreate the metrically disorienting surface.⁹



Example 5. Unbarred, un-beamed version of the opening of C. P. E. Bach, Symphony Wq. 183/1, mvt. 1.

While I will not provide a Mirka-style projective analysis in this article, arrows in Example 5 highlight how—once the lower layer of motion in eighth note enters—both quarter notes and half

pulsations, though more continuous ones. See: <https://www.cpebach.org/toc/toc-III-1.html>, accessed June 1st, 2023.

While this is at best circumstantial evidence, it might be indicative of exchanges of ideas about metric ambiguity in Bach's milieu.

⁹ This, of course, is similar to the presentation of the surface of Mozart's G-minor Symphony (K. 550) opening in Lerdahl and Jackendoff (1983, 74); see also Mirka's (2009, 49) survey of Koch's discussion of metric notation.

notes are established perceptually clearly. The “onset” of motion in eighth note is, in fact, the third attack point of the upper D5, which—in combination with the lower pulsations entering an eighth note later—starts this motion. What remains ambiguous is whether half notes are to be grouped into twos or threes, as I will explore below.

As we have witnessed through the harmonic reduction, the opening is tonally disorienting in addition to this unusual gesture of pulsations—perhaps mirroring Kirnberger and Schulz’s images. One might ask then, what is the *pitch-domain* equivalent of an undifferentiated stream of pulsations? It is a string of steps in pitch space, whether they occupy a diatonic space, the space of a triad, or some other space. Bach plays upon streams of stepwise scale degrees, or streams of motions within the space of a triad (that is, “stepwise” motion within the space of a triad) in several spots in this symphony. In this spirit, let us examine a couple of excerpts from Bach’s third and final movement in this symphony: at the conclusion of the finale (Example 6a), an undifferentiated ascent in pitch space through 13 diatonic pitches culminates in $\hat{3}$, which triggers a further ascent through $\hat{3}-\hat{4}-\hat{5}-\hat{6}$, that is, a Deceptive Cadence bass schema string, followed by $\hat{4}-\hat{5}-\hat{1}$, i.e., a Complete Cadence pattern (Gjerdingen 2007). The flourishes that follow these schematic bass gestures can also be conceptualized as occupying the space of a triad: in the case of the vi chord (B-minor chord) triggered by the schematic bass $\hat{6}$ (sounded as a unison), the sixteenth-note pulsations funneled through the space of the chord are deflected in pitch space by D5, creating an unusual pre-cadential hemiola; in the case of the final tonic flourish, the triumphant gesture is uninterrupted in pitch space and aligns more closely with the notated meter.¹⁰ Asterisks in Example 6a and throughout this article signify melodic accent stemming from registral extremes in a melodic line. The swarm of

¹⁰ Of course, one could make the argument that the D4 and D5 divide the penultimate measure and create a hemiola effect, but this is much more subtle in comparison with the groupings caused by the B3 and D5 registral boundaries.

undifferentiated, stepwise sixteenth notes culminating in the $\hat{3}$ that initiates the Deceptive→Cadence pairing can be taken as a metaphor for how scale-degree space and metric space constrain one another in eighteenth-century style: an undifferentiated series of scale degrees or pulsations may carry with it some level of ambiguity; what orients us are idiomatic ways in which scale degrees are organized metrically as typical schemata.

Deceptive Cadence 3 4 5 6 Complete Cadence 4 5 1

Example 6a. Symphony Wq. 183/1, mvt. 3, mm. 102–113.

The Deceptive→Complete Cadence pairing is strongly orienting within musical form, leading to decisive closure. What started as an atypical, non-schematic climb through pitch space and led to a melodic peak turns out to be $\hat{3}$ that activates a conventional cadential schema.

At the conclusion of the first reprise of his finale (Example 6b), Bach uses an unusually large pre-cadential hemiola. What creates the metrically dissonant grouping in the violins' layer is the fact that the motion up and down through the space of the A major triad is deflected at the registral

extremes of the arpeggiated line, C#4 and E5: these groupings compel us to shift our attention from the pre-established tactus (i.e., a whole measure of 3/8 in this case) and attend to a disruptive quarter-note tactus. Though a pre-cadential hemiola is well worn, Bach's trick here is unusual.¹¹

Complete Cadence 3 4 5 5 1

Example 6b. Symphony Wq. 183/1, mvt. 3, mm. 47–54(a).

Bach's pitch-spatial gestures, of course, create *melodic accents* based on the “sheer differences between ‘higher/lower’ tones” (Mirka 2009, 38). As Mirka observes, this is a usage of pitch as a

¹¹ An interesting compositional precedent to Emanuel Bach's pitch-spatial play is Johann Sebastian Bach's movement “Leite mich in deiner Wahrheit,” from the cantata BWV 150, “Nach dir, Herr, verlanget mich.” In Sebastian's chorus, a stepwise ascending gesture of four notes moves between the bass, tenor, alto, soprano, violin 2, and violin 1, traversing the space between B3 and to D6 over the course of 24 quarter-note beats, while cutting across a chordal textural layer.

statistical rather than *syntactical* parameter in Meyer's (1989, 1998) terms.¹² We can think about sheer spatial gestures such as the ones described above as a "statistical" use of pitch space whereas conventional schemata are "syntactical" ways of organizing pitch. The conclusion of the exposition of the first movement (Example 7a) expresses the layers of the notated meter through "statistical" motions in pitch space: after the almost-uniform ascent of m. 67, m. 68 descends stepwise from its initial highpoint. In other words, mm. 67–68 project the measure cycle into pitch space, putting a melodic accent on the highpoint on the downbeat of m. 68. The high-low patterns of mm. 69–70 project the quarter-note level, whereas m. 71 inverts the pattern into low-high while accelerating further, highlighting the contrast between strong and weak eighth notes within a quarter note. These gestures of highs and lows clearly project the meter, in contrast with the ambiguous opening. Note that—while mm. 67–68 feel like an arbitrary, non-schematic motion up and down through pitch space, mm. 71–72 clearly imply a cadential schema (recast into a chordal texture in Example 7b). Quite similarly to Examples 6a and 6b, this is a playful shift from pitch used closer to a statistical parameter (in the service of projecting metric dissonance or consonance) vs. pitch used syntactically, to create cadential closure.

¹² See also the comments in Mirka (2009, 45–46) and her example 2.9 (from Kirnberger and Schulz 1794, 101), regarding the affinity between pitch height and dynamic accents. In the first edition of Sulzer's encyclopedia, which would have been available to Bach at the time, this is at Kirnberger and Schulz (1774, 983). Melodic accents on highpoints will play a major role in my upcoming discussion of metric ambiguity.

[illegible]

Example 7a. Symphony Wq. 183/1, mvt. 1, mm. 67–72 (continued overleaf)

70

70

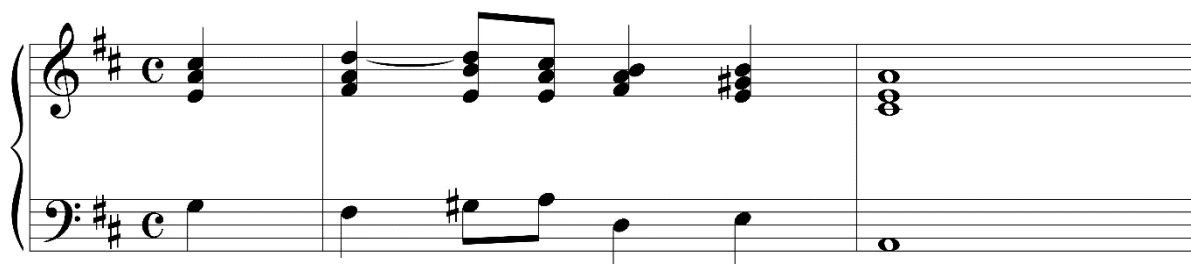
70

70

70

Complete Cadence: 1 4 5.....(!)

Example 7a (continued)



Example 7b. Textural recasting of the cadence of mm. 71–72.

3. *Opening Metric Ambiguity*

The opening sonic landscape of the symphony is indeed extraordinary. Accelerating syncopations in the first violins obstinately repeating the single note D5 interact with an arpeggiated, cascading gesture in the lower strings. Though the harmonic context is bright—with the first wave prolonging I, the second focusing exclusively on a vi sonority, and the third focusing on IV—the process of metric induction (Mirka 2009) offers ambiguity, captured above in Example 5. As suggested, the onset of eighth-note motion clearly establishes the quarter-note and half-note (tactus) levels. The remaining ambiguity is at the measure level: should we attend to groups of two or three half notes? Example 8a presents a reduction of mm. 1–7 in their original barring, whereas Example 8b re-bars the music in 3/2. Part of the problem here, of course, is that the subdivision into eighth notes (as well as quarter notes) is not articulated at the outset of the piece.¹³ While I will not engage here in Mirka-style projective analysis, one could paraphrase my observations about preference factors for the notated meter vs. 3/2 in terms of real-time projections.

¹³ Compare this to Mirka's (2009, 35) observations about her initial Haydn example for meter finding, example 2.1 (*Ibid.*, 34), the opening of Haydn's Quartet Op. 50, no. 6, mvt. 1.

(onset of eighth notes)

(=V4/2
of IV)

IV (6-----5)
3

V

Example 8a. C. P. E. Bach, Symphony Wq 183/1/i, mm. 1–7, author’s reduction of the orchestral score.

What factors support the notated meter, C? Our tendency to hear an initial accent and to impose binary regularity favor the notated meter from the outset in mm. 1–2, though the surface gives us few attack points to work with.¹⁴ One might experience the upwards arpeggio of the lower strings as an upbeat into the parallel highpoints at the downbeats of m. 3 and m. 4, which both carry a melodic accent of a high point.¹⁵ In addition, the most salient chord change in this passage is the one involving the resolution to a subdominant chord on the downbeat of m. 6, marking that

¹⁴ Mirka (2009, 39 table 2.1) lists these factors as “initial accent” and “binary regularity” (equivalent to Lerdahl and Jackendoff’s 1983 MPR 2, “strong beat early,” and MPR 10, “binary regularity”).

¹⁵ Lerdahl and Jackendoff’s (1983) parallelism rule is MPR1.

timepoint for our attention.¹⁶ The transformation of the tonic chord into a V4/2 of IV is more problematic for present purposes: should we hear a chord change on the fourth beat of notated m. 4? Should we pay attention to the (relatively) novel and salient C4 in the middle of m. 5? Under the notated meter, this is a point of incongruence, since the peak of m. 5's wave falls in its middle. Finally, the change to unison octaves on the downbeat of m. 7 supports the notated meter.

Though these are compelling factors in favor of the meter as notated, there are also some challenges: in particular, the (perceived) onset of eighth-note motion in the middle of m. 2 is incongruent with the meter. This suggests retrospectively interpreting the opening whole→half note pattern of the upper layer as a measure in 3/2. In addition, the indirect metric dissonances (in Krebs's 1999 terms) in the upper layer are disorienting: from m. 3, half notes are shifted by a quarter note, while quarter notes (in mm. 5–6) are displaced by eighth notes. This deprives the downbeats of mm. 3–6 of upper-layer attack points that might have made things clearer (and the downbeat of m. 5 has no attack point in either layer).

The re-barred Example 8b represents the meter as it may be heard initially. There are some attractive aspects to this reading, which I find more compelling subjectively.¹⁷ In the 3/2 hearing, the downbeat of re-barred m. 2 is the onset of (perceived) motion in eighth notes. Subsequently, the

¹⁶ As Temperley (2008, 318) observes, “not all harmonic changes are equal”: The root-preserving harmonic change from I to V⁷/IV is not as strong as the change to a IV chord in Bach's m. 6. As Mirka (2009, 39, 50–52) surveys, harmonic change is a most powerful preference factor and it is de facto subsumed under Lerdahl and Jackendoff's MPR5f and MPR8.

¹⁷ Since this piece is relatively obscure, it has provided me interesting opportunities for experiential conducting activities with students and colleagues in various teaching and presentation situations, with the task being to conduct the half-note tactus in either 2 or 3: in all cases, the ambiguity was visceral to the participant-attendees and seems to have elicited experiential engagement.

downbeats of re-notated m. 3 and m. 4 coincide with the melodic accent of the highpoints, clarifying the sense of 3/2. The downbeat of re-barred m. 5 is strongly emphasized through the unison gesture.

(onset of eighth notes)

mf

mf

(superfluous beat)

4

(main chord change to IV)

Example 8b. Re-barring of Example 8a. The final half-note beat of m. 5 does not exist in the original.

To be sure, the first arpeggiated peak, in re-notated m. 2, beat 2, is incongruent with the 3/2 hearing. Another drawback of this interpretation is the fact that re-barred m. 5 corresponds to original m. 7, which—in the actual score—includes only two half-note beats. In performance, however, there might be a tendency to extend this measure in a hesitant, fermata-like fashion—as in Anrew Manze’s delightful recording.¹⁸ The most salient chord change in the passage falls on the second half note of re-barred m. 4 in this hearing, again—weakening the 3/2 hearing and favoring

¹⁸ <https://www.youtube.com/watch?v=2PJ6TiWIw08>, accessed May 22nd, 2023. Of course, for an audience member in view of the conductor’s gestures, visual cues may provide a type of clarity that is absent with an audio recording.

the notated meter. Whatever its drawbacks, Example 8b outlines a plausible real-time hearing of this unusual opening.

The second, B-minor wave of arpeggios, is even more strongly suggestive of a 3/2 hearing, since it elongates a single harmony, vi, without any chord changes—see the original notation in Example 9a and a re-barred version in Example 9b. Thus, there are no chord changes present that might serve as metric cues.

Example 9a. Mvt. 1, mm. 8–14, author's reduction.



Example 9b. Re-barring of Example 9a in 3/2.

Moreover, successively rising highpoints on D4 and F#4 emerge on the downbeats of re-barred m. 10 and m. 11 (Example 9b), creating melodic accents. In other words, the second wave supports 3/2 more strongly than the first wave. The third and final wave (Example 10), on a IV sonority in mm. 15–18, is truncated, deviating from the pre-established pattern and leading to the surprising tutti texture of m. 19.

The events of m. 19 are remarkable (see Example 10): finally, the notated meter is established in our mind's ear. A powerful preference factor, chord changes, reinforces each downbeat from m. 19 to m. 24. Eighth-note pulsations continue, while various instruments highlight other subdivisions with simultaneous or *direct* displacement dissonances (after Krebs 1999): quarter-notes offset by an eighth note in the first violins (D2+1), as well as half notes offset by quarter notes in the flutes and oboes (D4+2, with 1=eighth note, see Example 10). Thus, displacements that were heard successively (or *indirectly*) at the outset are now sounded simultaneously in a tonally unsettling yet metrically clear tutti. These displaced layers interact with unsyncopated half notes in the horns as

well as unsyncopated whole notes in the continuo instruments. Thus, mm. 19–23 present a kind of “explanation” of the notated meter, though with a piling up of displacement dissonances.

As discussed earlier, the e-minor⁶ sonority of mm. 24–26 (see Example 10) serves as ii^6 in the home key at the end of the tonal disruption. In addition to the tonal clarification, the zig-zag of mm. 24–26 projects the metric grid through melodic accents: each quarter note is subdivided into a high-low pattern, while a two-measure unit emerges in mm. 24–25 and contrasted with a 1-measure gesture in m. 26. Now no displacement dissonances or wrinkles are left: the zig-zaggy pattern projects the notated meter clearly and audibly through pitch space.¹⁹

¹⁹ An alternative interpretation would be that E pitches are onsets of triadic patterns such that E6, E5, E4, E4, E5 create a 3+3+2+2+2 subdivision in mm. 24–26, that is, a hemiola effect. Perhaps this interpretation is supported by the cessation of chord changes, which had created clarity in the tutti texture. While I can see the merit of such a hearing, I find it more plausible to maintain the notated-and-heard meter that had been established forcefully from m. 19.

15

ff

D4+2

15

ff

ff

D2+1

15

ff

ff

ff

15

ff

7 5

6 4

ff

Example 10. Symphony Wq. 183/1, mvt. 1, mm. 15–26 (continued overleaf)

21

D4+2

a2

D2+1

p

pp

p

p

p

21

7
5

5

4

6

Example 10 (continued)

The image displays a musical score for the piece "Passo Indietro" by Giuseppe Verdi. The score is written for piano and violin. The piano part is in the upper system, and the violin part is in the lower system. The key signature is one sharp (F#), and the time signature is 4/4. The score is divided into two systems, each containing four measures. The piano part features a melodic line with a "single-note idea" and a bass line with a "single-note idea". The violin part features a melodic line with a "single-note idea" and a bass line with a "single-note idea". The score is marked with a forte (f) dynamic. The tempo is marked "Allegretto". The score is numbered 27 at the beginning of each system.

Example 11. Symphony Wq. 183/1, mm. 27–34 (continued overleaf)

31

(sus. chain)

Prinner

6 5 4 3 2

I: HC MC

31

9 6 7 —

31

9 6 7 — 9 7 — 7 5 5 unis.

4 3 7 1 5

Detailed description: The image shows a musical score for a piece titled 'Example 11 (continued)'. It consists of several systems of staves. The first system has a single staff with a treble clef and a key signature of one sharp (F#), starting at measure 31. The second system has two staves: the top staff is in treble clef with a key signature of one sharp, and the bottom staff is in treble clef with a key signature of one sharp. The third system has four staves: the top staff is in treble clef with a key signature of one sharp, the second staff is in treble clef with a key signature of one sharp, the third staff is in bass clef with a key signature of one sharp, and the fourth staff is in bass clef with a key signature of one sharp. The fourth system has two staves, both in bass clef with a key signature of one sharp. The score includes various musical notations such as notes, rests, and fingerings. Performance instructions like '(sus. chain)', 'Prinner', and 'I: HC MC' are present. The piece concludes with a 'unis.' instruction.

Example 11 (continued)

So far, the piece has told an incongruous “story” of ambiguity and clarity in meter and tonality. Yet there is one more analytical story—that of galant schemata within form (e.g., Byros 2015, Caplin 2015). In terms of galant schemata, the twice-repeated Passo Indietro schema of mm. 27–28 (Example 11) sounds premature, since this schema typically precedes the final cadence of a formal section.²⁰ The Passo Indietro, however, does help the opening I-vi-IV harmonic model become more like an outline of a galant Romanesca schema writ-large: it completes the “missing” descent to bass $\hat{3}$. Measures 31–34 (see Example 11) put the piece back in its track as far as schemata-within-form are concerned: they clearly outline a melodic descent through a Prinner→HC succession or $\hat{6}-\hat{5}-\hat{4}-\hat{3}-\hat{2}$, culminating in a I: HC MC that clarifies that the piece is no longer an unusual orchestral fantasy but has rather become a typical sonata-form movement.²¹ Example 12 summarizes the metric and tonal clarity vs. ambiguity and schematic typicality vs. atypicality in the first part of the exposition: only at m. 31 do we have a coincidence of metric and tonal clarity with a typical schema-within-form. (Example 12 also includes an additional column regarding metric states, to be discussed later). Of course, after the improvisatory opening arpeggiations and unusual tonal and metric events, the extremely conventional Prinner-HC outline that becomes a conventional signpost, a I: MC HC, may sound contrived. Whatever our aesthetic reaction, it is interesting to revisit Bach’s (1762) advice about rational deceptions in a fantasy, which should not be used excessively. In this spirit, convention is restored here.

²⁰ In Gjerdingen’s words (2007, 167), “a descent in the bass from $\hat{4}$ to $\hat{3}$ often preceded a stronger cadence.” See also [redacted].

²¹ The tonally open ending of the first two movements and the unusual tonal relation of the inner movement (key of E-flat major, i.e., bII) also gives fantasy-like flavor to this symphony. Nevertheless, the obscure opening of the first movement is probably the most fantasy-like aspect of the work.

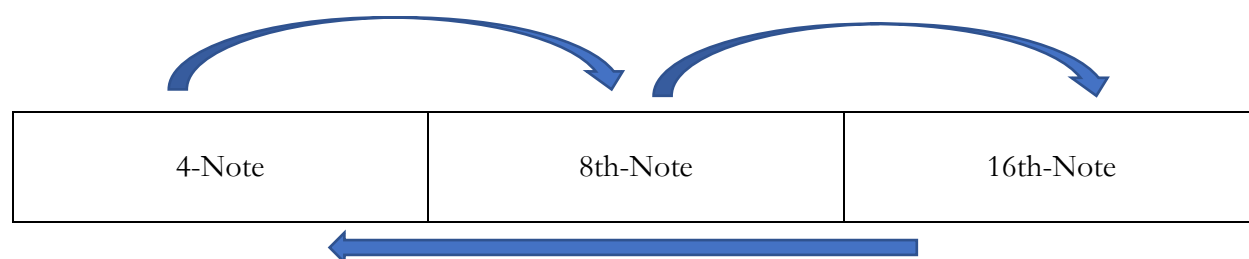
Parameter Measure numbers	Perceived meter	Tonality	Harmonic/contrapuntal schemata within form	Metric state (lowest subdivision available)
1–18	<i>Obscure</i> (3/2 or notated C)?	Clear, D major	Galant Romanesca, opening (typical)	8 (eighth notes)
19–26	Clear	<i>Obscure</i>	A minor (v)→B minor (vi) implications (<i>atypical</i>)	8 (eighth notes)
27–30	Clear	Clear, D major	Passo Indietro*2 (<i>atypical</i>)	16 (sixteenth notes)
31–34	Clear	Clear, D major	Prinner→I: HC MC (=6̂–5̂–4̂–3̂–2̂, typical)	16 (sixteenth notes)

Example 12. Summarizing the interactions of meter, tonality, harmonic/contrapuntal schemata, and metric states in mvt. 1, mm. 1–34

4. *S Zone: Slow State and Monothematic Repetitions*

Example 12 contains one more column, representing metric states in terms of the fastest-available subdivision at any given moment. As we can see, mm. 1–26 are confined to eighth notes as the fastest-available subdivision, while m. 27 brings in faster surface rhythm, conventionally setting up the path towards the MC. In contrast, the S zone starting in m. 35 (Example 14) moves us to the metric state with the slowest pulse, in which quarter notes are the fastest available pulsating layer. Given Hepokoski and Darcy's (2006) detailed description of the MC and surface-rhythmic activity surrounding it, the trajectory of metric states is conventional: the piece starts at an intermediate speed of surface activity, accelerates towards the MC, and decelerates with the onset of S. Yet in this unusual piece, the metric states act as compositional constraints: the motion into the quarter-note-as-lowest-subdivision state virtually eliminates shorter note values, with the rare exceptions of written-out embellishments (in m. 44 and m. 47), which do not amount to a consistent layer of pulsations. We can conceptualize these states as nodes within the very simple metric space of Example 13. The motion across the MC is thus a leap from the fastest to the slowest moving state

within the movement. Thus, the opening arpeggios were a hesitant way of introducing the baseline metric state (the middle one, with eighth notes as the fastest pulsation present).



Example 13. Possible “steps” and “leaps” between metric states (based on the lowest-present pulsations, be they quarter notes, eighth notes, or sixteenth notes).

Though the S materials sound fresh, the piece exhibits an intricate web of compositional ideas previously presented in the first part of the exposition (Example 14). The second oboe in mm. 35–36 articulates the single-note idea with which the first violins opened the piece: the oboes subsequently enter a chain of 2–3 suspensions that supports the notated meter through the age-old association between suspensions and downbeats. Though the oboes initially repeat the opening single-note idea, they get stuck on the chain of suspensions.²² The single-note compositional idea from the opening (in the first violins) is thus translated to the solo bassoon, whose melodic accents in mm. 35–44 create displacements akin to those of the opening first violins’ single-note idea, now embedded within a slower metric state. In m. 44, the violins enter with the single-note idea and accelerate in a syncopated way to the maximal extent possible in the prevailing state. In other words, after the bassoon’s rhythmic foreshadowing, the idea is reiterated by the violins. In addition, the E6 highpoint of the flutes in m. 45 foreshadows the violins’ syncopation, since it is on the second beat;

²² Both the single-note idea and the suspension-chain idea (in a 7-6 guise) are foreshadowed in the oboes and horns in mm. 27–30 and 31–32, respectively. The former idea is also reinforced stereophonically through the first and second violins in alternation in mm. 27–30 (refer back to Example 11).

the subdivided beat 2 of m. 47, which stands out from its surroundings, coincides with the violins' syncopation. Thus, the interplay of the flutes and violins in mm. 44–48 continues to play upon the single-note idea and its metric dissonances.

The monothematic relations do not end here, since the events of mm. 35–44 also echo preceding harmonies. Construed as melodic galant schemata, the events of mm. 37–44 are a Prinner → Converging Cadence soprano outline ($\hat{6}-\hat{5}-\hat{4}-\hat{3}-\hat{2}-\hat{1}-\hat{7}$) in the key of V. The onset of the Prinner as a point of arrival in m. 37 coincides with an absolute-pitch harmonic echo of preceding events: monotonal I, vi, and iii (D major, b minor, f# minor sonorities) with D–B–F# bass outline (monotonally $\hat{1}-\hat{6}-\hat{3}$), that is, an echo of the disrupted opening Romanesca. A competing schema, a Fonte sequence in mm. 40–41, creates an even denser web of echoes: the bass $\hat{7}-\hat{1}$ of the second wing of the Fonte (in m. 41) echoes the D#–E of the tonal disruption (mm. 19–20). In other words, though events are appropriately shifted to the key of V, monotonal harmonies are used as motives.

35

Example 14. Symphony Wq. 183/1, mvt. 1, mm. 35–48 (continued overleaf)

42

42

Conv.
Cad.

2 1 7

*

* (accented by the subdivision)

tr

(single-note idea)

42

p

p

42

D4+1 Conv. Cadence

*

4# 5

42

Example 14 (continued)

In mm. 56–62 within S (Example 15), Bach takes the mysterious opening six hammer strokes or pulsations and sculpts from them a tuneful bass line instantiating the Fenaroli schema (see the bass layer in mm. 56–57).²³ The clarity of the Fenaroli has a lot to do with the fact that the meter of the piece had been perceptually established by now, as well as by the conventional outline of the Fenaroli. In addition, the Fenaroli’s built-in dominant-tonic harmonic alternations (=chord changes) reflect back on the opening hammer strokes and “disambiguate” them. This is added to the web of relations within the movement. It also participates in the game of undifferentiated motion through a space (meter or pitch) vs. a sculpted schema, or the statistical-to-syntactic trajectory: the inchoate mass of the opening pulsating gesture is molded into a conventional, tuneful schema, which is, moreover, embedded in its typical position within sonata form (S zone).²⁴ Musical order has thus been imposed on the artisan’s opening hammer strokes.

²³ Note that one could apply Brody’s (2016, 113) logic of *schema* (parametric correlation) vs. *matrix* (parametric independence) to the embedding of diatonic scale degrees within eight descending sixteenth-note runs. Gestures framed by $\hat{2}$ (representing Gjerdingen’s High- $\hat{2}$) and $\hat{2}$ or $\hat{4}$ and $\hat{4}$ an octave apart—such as those of m. 57, m. 59, and m. 61 in the violins—seem more conventional in the galant idiom compared to such flourishes bounded by different contextual scale degree (cf. [redacted]).

²⁴ For a review of the Fenaroli’s main features, see Gjerdingen (2007, 462).

Example 15. Symphony Wq. 183/1, mvt. 1, mm. 56–62 (continued overleaf)

(Comma) Cadence (Evaded)

Example 15 (continued)

5. *Development Section: Virtue in a Strategy Destined for Extinction?*

My attention so far has been primarily on rhythm and pitch, though other considerations have crept in. In discussing the development, I would like to shift my focus to large-scale formal strategies. Bach's development section is a stable rotation (in Hepokoski and Darcy's 2006 terms), revisiting the events of the exposition almost in order. Given our enculturation in high-classical sonata norms, such a strategy may seem pedestrian to us: we expect a stormy, unstable section, developing and even liquidating the materials of the exposition. Greenberg notes Bach's use of such a stable rotational strategy, in which "the entire development section is constructed along the lines of a rotation" (2022, 135). Though he eschews a simplistic teleological view of evolution towards the high-classical sonata form, he observes that "excessive repetition" is the weakness that "eventually led to the abandonment of this formal plan as a standard option" (2022, 138). In other words, this strategy did not survive in the gradual evolution from binary forms into what we now think of as sonata form.

	Opening Arpeggio Waves		"D"→"b" Interpolation		"MC- S" blend	→To cadence in vi (conventional tonal destination)	Retransition
Development	72–89	90–93	94–97	98– 101	102– 107	108–122	122–135
Exposition Equivalent	1–18	Compare 91–93 to 27–29		31– 34		56–71	
Metric state (fastest subdivision available)	8 th	16 th	16 th	16 th	4/16 th	16 th	4 notes (8 th notes from m. 131, 16 th notes from m. 133 as tremolo)

Example 16. Comparison of the exposition and development

Example 16 provides a synoptic view of the parallelisms and differences between the exposition and development. In addition, it lists the metric states in terms of the fastest available pulsating subdivisions. What might be a compositional advantage for this strategy—a formal trait on the verge of extinction—in the case of this idiosyncratic symphony? A couple of possible explanations come to mind. First, there are no notated exposition repeats (or any kind of repeats) in this movement. Thus, the developmental rotation makes up for the missing exposition repeat. Second, the chief point of deviation from the course of events of the exposition, which I have marked as “MC–S” blend in Example 16, is an opportunity for a stark juxtaposition between the two extreme metric states of the movement. Thus, the deviation from the stable rotational logic highlights the stark contrast between the polar extremes of metric states in the movement. Bringing in an “MC–S” blend rather than repeating mm. 35–48 (Example 14) also prevents the energy loss of the onset of S, maintaining momentum. In other words, this is yet another solution to the problem of stability in the developmental rotation. Finally, the retransition is an opportunity for further metric play and acceleration between metric states.

6. *Conclusion*

Bach’s rational metric, rhythmic, harmonic, pitch-spatial, and formal manipulations in this movement have required an ad-hoc, multi-parametric analysis. Of course, those who favor a different combination of tools, or even the same ones, might come up with divergent interpretations of this captivating symphonic puzzle. How should we understand Bach’s symphony? As an implicit reflection on matrices, spaces, and schemata? As an exercise in moving between harmonic and metric ambiguity? As an experiment with interleaving metric states and gradually unravelling metric dissonances? As a way of molding the opening, prelude-like and undifferentiated arpeggio strokes into a tuneful Fenaroli schema? As an exercise in moving from pitch as a statistical parameter whose

ups-and-downs articulate metric dissonance or consonance, towards pitch as a syntactic parameter that is associated with form-defining schemata and cadences? Whatever the case, Bach's symphony invites close listening and reflection, connecting real-time intuitions and impressions with offline speculation and score study. I have indulged here in both types of engagement, with the hopes that readers would be encouraged to experience this remarkable piece in these and many other ways.

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